

## Problem #1:

1. The answer is yes I think if you combine all three methods. I had Claude help me write the script for that.

## Problem #2:

1.
  - a. Supervised Learning Classification: Image classification.
  - b. Supervised Learning Regression: Keras regression block for continuous data prediction like temperature forecasting.
  - c. Unsupervised learning anomaly detection: Detecting abnormal vibrations.
  - d. Unsupervised learning visual anomaly detection: Inspecting PCBs for defects
2.
  - a. Input layer: defines the shape of incoming data
  - b. Dense: Core computation layer for feature extraction and classification
  - c. Reshape: Alters the tensor dimensions, for example flattening sequences
  - d. Flatten: This converts multidimensional data into 1D before Dense Layers
  - e. Dropout: Randomly disables neurons during training to reduce overfitting and over reliability
  - f. 1D Convolution and Pooling: For temporal or sensor signals
  - g. 2D convolution and Pooling: For image data such as image classification and object detection
  - h. Output Layer: Provides model predictions
3.
  - a. Quantization
  - b. Unoptimized
  - c. EON Compiler
4.
  - a. Images: black forest labsflux-pro synthetic image generator
  - b. Images: DALLIE-3 Synthetic Image generator
  - c. Audio: Whisper Synthetic Voice Generator